

[EA&B] When Blind Feature Removal Mitigates Tabular Leakage

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Abstract

Feature contamination can make offline tabular evaluation optimistic, but an invalid field need not be easy to locate or useful to every learner. This distinction also complicates governance: removing highly associated fields may repair simple contamination while missing leakage encoded by time, entity, or source structure. We present LeakBench-Tab, a frozen experimental benchmark that separates construction validity (C), statistical detectability (D), and model exploitability (X), then evaluates leakage removal under matched feature budgets. The controlled measurement matrix contains 27,500 paired cells across 20 panel tasks, 11 mechanisms, five strengths, five seeds, and five model families. In this registry, simple mechanisms produce 0.164 more paired AUROC harm than structured mechanisms (95% CI [0.150, 0.178]). A separate oracle-isolated governance experiment compares blind training-side mutual information (P3) with the mean of 20 matched random-removal realizations (P2). At a 20% encoded-column budget, the P3–P2 strict-distance-reduction contrast is +0.043 for logistic regression (95% dataset-cluster bootstrap CI [+0.004,+0.078]), +0.055 for random forest ([+0.025,+0.082]), and +0.056 for LightGBM ([+0.028,+0.082]). Mechanism-level analysis shows that the structured average mixes near-zero-gap mechanisms with the positive M09 source-leakage counterexample. When the eight M09 indicators are charged as one semantic group, M09 remains positive (+0.109, [+0.054,+0.158]), but the recomposed overall interval crosses zero (+0.040, [-0.003,+0.077]). Five fixed natural cases provide mixed descriptive support: four favor P3 and NYC311 favors random removal. Blind scoring can therefore mitigate leakage in this controlled panel, but the result depends on repair opportunity, representation cost, and application context.

Keywords

data leakage, tabular benchmarks, experimental analysis, feature governance, reproducible evaluation

1 Introduction

Tabular prediction pipelines routinely inherit fields whose values are only available after the prediction event, are derived from the target, or summarize future entities and outcomes. Such contamination can invalidate evaluation and distort model comparison even when the learning algorithm and held-out split are otherwise conventional [8, 9]. The practical response is often to remove suspicious fields, but this response combines three different questions: whether a field is semantically valid at prediction time, whether an oracle-blind statistic can locate it, and whether a learner exploits it enough to change measured performance.

Existing leakage studies establish the importance of prediction boundaries, scientific validity, and domain-specific label leakage [4, 14]. Detection work also spans code analysis and empirical contamination audits [11, 16]. These perspectives do not by themselves determine whether a deployable, budget-limited removal policy will repair an affected tabular pipeline. A detector can rank target proxies highly yet fail on future aggregation or entity structure. Conversely, a semantically invalid field can be hard to rank but still inflate a learner’s score.

This gap matters because leakage response is an intervention, not merely a diagnosis. Removing a field changes the representation on which the learner is fitted and may discard legitimate predictive information along with the contamination. Comparing a removal policy only with the unmodified table confounds two effects: locating invalid information and reducing model capacity. Comparing it only with an oracle mask instead asks whether the policy is perfect, not whether its ranking is useful under realistic constraints. A matched random policy supplies the missing negative control: it estimates how much movement toward a strict reference is obtained merely by deleting the same number of encoded fields.

A credible benchmark for this question must preserve several separations at once. Semantic invalidity must be declared before examining feature scores; diagnostics must be fitted without the contamination mask; strict, permissive, and governed learners must share task content and test rows; and uncertainty must respect the task hierarchy rather than treating repeated mechanisms, strengths, seeds, or models as independent datasets. These requirements turn a seemingly simple feature-ranking comparison into a protocol-design problem.

This paper treats leakage evaluation and governance as an experimental analysis problem rather than proposing a new feature selection algorithm. LeakBench-Tab crosses eleven audited mechanisms with controlled panel tasks, strengths, injection seeds, and learner families. It records three distinct quantities: construction validity (C), statistical detectability (D), and model exploitability (X). We then evaluate a representative blind policy, training side mutual information removal, against a matched cost random baseline. Policy selection is isolated from the contamination mask; the mask is used only after selection to evaluate recall and legitimate field retention.

Our analysis asks three questions. **RQ1** characterizes how D and X vary across contamination mechanisms. **RQ2** tests whether blind feature removal improves a governed model’s distance to the strict reference under matched budgets. **RQ3** identifies the mechanism families for which the policy advantage persists or fails. The results show a consistent overall P3 advantage in the logistic regression panel, including at small budgets, but also expose a sharp boundary: simple and boundary mechanisms drive the gain, whereas structured mechanisms

show no reliable advantage. The contribution is therefore an empirical map of strengths and failure conditions, not a claim that mutual information is a general leakage detector.

Contributions. The paper makes four bounded contributions. First, it operationalizes leakage as a three-axis measurement problem: construction validity is supplied by the prediction-time contract, detectability is conditional on a blind diagnostic, and exploitability is conditional on a learner. Second, it provides a frozen factorial registry that varies task archetype, contamination mechanism, strength, injection seed, and learner while preserving paired strict and permissive evaluation. Third, it connects measurement to action through an oracle-isolated, matched-cost governance experiment with explicit localization and legitimate-retention outcomes. Fourth, it reports both the positive result and its failure boundary: blind MI removal improves the LR panel overall, but the gain is not reliable for structured contamination. The artifact binds each paper-facing estimate to canonical cells, protocol manifests, analysis code, and content hashes.

2 Problem Formulation

2.1 Prediction Boundary and the C/D/X Axes

Let $q = (t, m, a, s)$ index a controlled cell by task t , contamination mechanism m , strength a , and injection seed s . Its encoded feature universe is $\mathcal{F}_q$, and the construction contract identifies the contaminated subset $\mathcal{L}_q \subseteq \mathcal{F}_q$. The strict and permissive views are therefore

$$\mathcal{F}_q^{\text{strict}} = \mathcal{F}_q \setminus \mathcal{L}_q, \quad \mathcal{F}_q^{\text{full}} = \mathcal{F}_q. \quad (1)$$

Each task declares an information boundary at prediction time t_p . For encoded field j , construction validity is

$$C_{qj} = \mathbf{1}[j \in \mathcal{L}_q] = \mathbf{1}[j \text{ is unavailable or target-derived at } t_p]. \quad (2)$$

C_{qj} is fixed by task semantics and mechanism construction. It is not inferred from association, attribution, or downstream performance.

An oracle-blind diagnostic assigns score s_{qj} without reading C_{qj} or the test rows. We measure localization with average precision and normalize it by the contaminated field prevalence π :

$$\pi_q = \frac{|\mathcal{L}_q|}{|\mathcal{F}_q|}, \quad D_q = \frac{AP_q - \pi_q}{1 - \pi_q}. \quad (3)$$

The primary diagnostic is mutual information computed on training rows with a fixed random state. D is conditional on the diagnostic and representation.

For learner ℓ , exploitability is the paired AUROC difference

$$X_{q\ell} = A_{q\ell}^{\text{full}} - A_{q\ell}^{\text{strict}}, \quad (4)$$

where ℓ indexes the downstream learner and A denotes test AUROC. The strict view removes construction-labeled contaminated fields; the permissive view retains them. Positive X measures performance inflation under this protocol. An interval containing zero is reported as an imprecise or unsupported directional effect, not as evidence of equivalence.

2.2 Governance Objective

For a policy r and removal budget k , let $\mathcal{R}_{q,r,k} \subseteq \mathcal{F}_q$ be its selected removal set. The governed feature view and its matched-cost constraint are

$$\mathcal{F}_{q,r,k}^{\text{gov}} = \mathcal{F}_q \setminus \mathcal{R}_{q,r,k}, \quad |\mathcal{R}_{q,r,k}| = k. \quad (5)$$

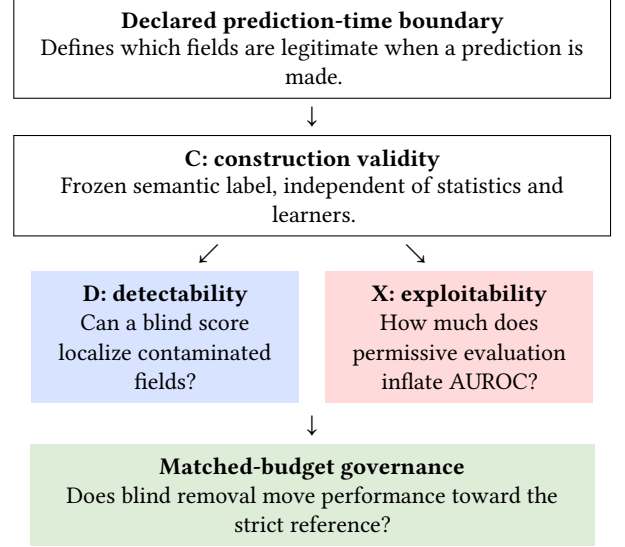


Figure 1: Simplified measurement-to-governance architecture. C defines the semantic target. D and X are separate empirical measurements. Governance is evaluated against the strict reference under matched removal cost. A more detailed system architecture is intentionally deferred.

Let A_s , A_f , and A_g denote strict, full, and governed test AUROC for a fixed q and learner. We define strict distance reduction (SDR) as

$$\text{SDR}_{q,r,k} = |A_{f,q} - A_{s,q}| - |A_{g,q,r,k} - A_{s,q}|. \quad (6)$$

Positive SDR means governance moves the evaluated learner closer to the strict reference. This symmetric distance avoids treating negative residual harm as automatic success. We also report contaminated field recall and legitimate retention:

$$\text{Rec}_{q,r,k} = \frac{|\mathcal{R}_{q,r,k} \cap \mathcal{L}_q|}{|\mathcal{L}_q|}, \quad (7)$$

$$\text{Ret}_{q,r,k} = 1 - \frac{|\mathcal{R}_{q,r,k} \setminus \mathcal{L}_q|}{|\mathcal{F}_q \setminus \mathcal{L}_q|}. \quad (8)$$

These outcomes are evaluation quantities and can read C after a policy has selected fields; non-oracle policy selection cannot.

2.3 Evaluation Conditions

Four conditions distinguish the benchmark from ordinary feature selection. *Semantic precedence* requires $\mathcal{L}_q$ to be determined by the prediction-time contract rather than by a score threshold. A feature can be invalid with low marginal association, or valid with high association; neither case is contradictory. *Oracle isolation* requires the selection map $\mathcal{R}_{q,r,k} = \phi_r(X_q^{\text{train}}, y_q^{\text{train}}, k)$ to exclude $\mathcal{L}_q$ for non-oracle policies. The mask enters only after $\mathcal{R}_{q,r,k}$ is frozen, when recall and retention are evaluated.

Cost matching holds the encoded-field removal count fixed within each comparison. This is important because one-hot expansion makes a semantic field and an encoded column different units. The present experiment governs encoded columns: removing all eight source indicators for M09 costs eight units, while removing a single numeric target proxy costs one. The paper therefore does not equate a nominal percentage with a constant semantic intervention across mechanisms. Instead, it exposes both the nominal fraction p and the realized integer $k_q(p)$.

Paired reference evaluation keeps the task, split, mechanism, strength, injection seed, and learner seed fixed when strict, full, and governed AUROC are compared. SDR is bounded by $[-1, 1]$ because it is a difference of two AUROC distances. Positive SDR indicates movement toward the strict score, but not recovery of the strict feature set: a policy may reach a similar AUROC after removing the wrong fields. Recall and retention are reported alongside SDR precisely to reveal that ambiguity. Likewise, negative SDR identifies movement away from the strict reference but does not by itself distinguish residual contamination from over-removal of legitimate signal.

These conditions define the scope of a governance claim. A policy is useful in this benchmark only when it improves the paired distance under equal removal cost and its localization trade-off remains visible. The formulation does not assert that the strict model is optimal for a business objective, only that it is the evaluation reference consistent with the declared information boundary.

3 Benchmark and Experimental Design

3.1 Controlled Registry

The controlled registry contains 20 deterministic panel tasks spanning five data-generating archetypes. Each task has a chronological 60/20/20 split and separates generator seeds from injection seeds. Eleven mechanisms cover direct target copies (M01), noisy proxies (M02), nonlinear transforms (M03), post-outcome aggregates (M04), temporal look-ahead (M05), redundant clusters (M06), sparse subgroup leakage (M07), entity leakage (M08), source leakage (M09), mixed legitimate and contaminated components (M10), and graph-mediated leakage (M11). Categories are assigned by construction: M01, M02, M06, and M10 are simple; M03, M07, and M11 are boundary mechanisms; M04, M05, M08, and M09 are structured. Categories are not revised after observing results.

The five base archetypes are linear, interaction, nonlinear, sparse, and drifting binary prediction tasks. Each contains entity and time structure and between 12 and 24 original numeric fields. Clean signal combines row-level covariates with entity effects and a periodic time component; the archetype then changes the functional form rather than the evaluation contract. Four tasks instantiate each archetype. Task generator seeds are deterministic functions of the panel identifier, so changing an injection or learner seed cannot regenerate the clean table. All tasks are ordered by timestamp before the 60/20/20 split, and injection is verified not to alter the frozen row indices.

This panel is designed for controlled variation rather than realism by volume. It supplies repeated task structure in which semantic contracts and injected features are exactly auditable. The five natural cases introduced below serve a different purpose: they test whether boundary-specific contamination remains visible in selected real tables, but they do not replace the controlled panel as the inferential unit.

The core matrix crosses 20 tasks, 11 mechanisms, five strengths, five injection seeds, and five learners, yielding 27,500 paired cells. The learners are logistic regression, random forest [1], CatBoost [13], LightGBM [10], and TabM [6]. Preprocessing and diagnostic fitting use training data only. The design evaluates conventional linear, bagged-tree, boosted-tree, and neural tabular families without treating model family differences as causal capacity effects.

3.2 Mechanism Families and Strength Semantics

The registry varies how invalid information becomes available to a learner. The *simple* family contains fields whose encoded columns directly expose or redundantly repeat target information. M01 copies the current label with a strength-dependent flip rate; M02 adds noise to a centered target proxy; M06 creates a correlated cluster whose size increases across strength levels; and M10 pairs one legitimate copy of an original field with one contaminated target proxy. M10 is deliberately mixed: the strict view removes only its contaminated component and retains the legitimate component, preventing the oracle reference from benefiting merely by deleting an entire injected group.

The *structured* family makes validity depend on relations not represented by a single contemporaneous field. M04 aggregates labels from a strictly future window; M05 introduces a configurable look-ahead offset; and M08 computes a shrunken, strictly future label rate within entity. All three exclude the current row's own label. M09 assigns one of eight source categories with outcome-dependent source probabilities and represents the source as a complete one-hot block. It does not use target-rate encoding, but the source identity is deployment-invalid under the declared construction. Consequently, its detectability and removal cost are representation-conditional.

The *boundary* family stresses cases between those poles. M03 multiplies a target-derived signal by the side of a threshold on a legitimate field, hiding marginal association behind an interaction. M07 exposes a target proxy only within a sampled subgroup, with subgroup prevalence varying by strength. M11 distributes target-derived signal across a connected component whose size also varies with strength. These mechanisms are not claimed to enumerate all forms of leakage. They are controlled probes for directness, sparsity, interaction, redundancy, time, entity, source, and group structure.

Strength labels S1–S5 correspond to values 0.2, 0.4, 0.6, 0.8, and 1.0, but the operational parameter is mechanism-specific. Strength scales signal for direct proxies, changes subgroup coverage for M07, and changes group size for M06 and M11. It therefore defines an ordered intervention within a mechanism, not a common physical scale across mechanisms. Cross-mechanism comparisons average the frozen five-level profile; they do not interpret S3 in one mechanism as quantitatively equivalent to S3 in another.

3.3 Diagnostics and Learner Protocol

The primary diagnostic assigns each encoded field a univariate MI score using training rows and a fixed random state. Localization is evaluated after scoring with normalized average precision, which adjusts for the mechanism-specific prevalence of contaminated encoded columns. Average precision is appropriate for this imbalanced ranking problem because most fields are legitimate and the relevant operating region is concentrated near the top of the ranking [15]. We also record top-five contaminated-field recall as an operational summary, but the normalized AP metric supplies D in the C/D/X analysis.

A frozen sensitivity suite evaluates three alternatives: absolute Pearson correlation on training rows, absolute coefficients from standardized logistic regression, and validation-set permutation importance from a training-fitted random forest. None receives the contamination mask before producing scores. The four-method suite contains 22,000 successful diagnostic

cells. Because the protocol did not freeze simultaneous paired method-comparison intervals, these results diagnose method dependence but do not support an inferential ranking of detectors. In particular, a retrospectively selected best method is an optimistic labeled-benchmark summary, not a deployable selector.

The downstream panel fixes five named implementations. LR uses standardized features and logistic regression; RF uses 250 trees; CatBoost and LightGBM use validation-only early stopping; and TabM uses the official pinned package with 32 ensemble members, training-fitted standardization, deterministic execution, and validation-loss checkpointing. Test labels are used only to compute final AUROC. Missing packages or accelerators fail a cell rather than triggering a fallback under another model identity. The completed canonical matrix contains all 27,500 planned model cells, so no failed-cell exclusion enters the reported means.

3.4 Matched-Cost Governance Policies

The governance experiment uses the same 5,500 combinations of task, mechanism, strength, and seed. Logistic regression, random forest, and LightGBM are the downstream learners. P0 retains all fields. P1 removes every field labeled contaminated by construction and serves only as an oracle reference. P2 removes k fields uniformly at random under 20 frozen governance seeds. P3 removes the k fields with the largest training side mutual information scores. More formally, if $\omega \in \Omega$ denotes one of $|\Omega| = 20$ frozen random-policy realizations and I_{qj}^{train} denotes the training-only MI score, then

$$\mathcal{R}_{q,P2,k}^{(\omega)} \sim \text{Unif}\{R \subseteq \mathcal{F}_q : |R| = k\}, \quad (9)$$

$$\mathcal{R}_{q,P3,k} = \text{TopK}_{j \in \mathcal{F}_q} (I_{qj}^{\text{train}}; k). \quad (10)$$

For a nominal budget p of 1%, 5%, 10%, or 20% and $d_q = |\mathcal{F}_q|$ encoded fields, both policies use

$$k_q(p) = \max(1, \text{round}(pd_q)). \quad (11)$$

Thus the comparison satisfies the cellwise cost identity

$$|\mathcal{R}_{q,P2,k}^{(\omega)}| = |\mathcal{R}_{q,P3,k}| = k_q(p). \quad (12)$$

P2 and P3 therefore remove exactly the same number of fields for each key and budget. The primary contrast uses $\Delta_q = \text{SDR}_{q,P3} - |\Omega|^{-1} \sum_{\omega \in \Omega} \text{SDR}_{q,P2}^{(\omega)}$ before dataset-cluster resampling. Nominal budgets that collapse to the same integer k reuse the same fitted result and retain both labels in the ledger.

The clean runner loads frozen bundles, verifies bundle and split hashes, and does not inject, split, or mutate data. Non-oracle selectors never receive the contamination mask. Strict and full baselines are shared only within an identical task, mechanism, strength, and seed key. All 55,000 policy cells completed successfully across 5,500 keys, with no duplicate run identifiers.

3.5 Natural-Case Audit

Five selected public or locally mirrored tables instantiate concrete prediction boundaries: call completion for Bank Marketing, loan origination for Lending Club, scheduled departure for U.S. flights, inspection completion for Chicago food inspections, and complaint creation for NYC 311. For each task, the strict view removes fields declared unavailable at that boundary and the permissive view retains them. The harm audit crosses four CPU learner families with three seeds, producing 60 paired cells. A separate LR governance sensitivity uses the 20% retained-feature

budget, three training seeds, and 20 P2 governance seeds, producing 315 policy cells. The five cases are selected rather than sampled, so both analyses remain descriptive fixed-case evidence.

Categorical preprocessing is fitted on training rows. Categories absent from training map to a reserved unknown value, missing categories receive a separate value, and date strings that had previously been globally encoded are dropped. This policy prevents validation or test vocabularies from defining the training representation. The case selection is intentionally heterogeneous in table width and contamination count, but it is not random and does not define a population of application datasets. One Lending Club source remains a local mirror whose redistribution license is unresolved; the paper reports its boundary-specific result but excludes the raw file from the public artifact.

3.6 Freeze and Amendment Discipline

Confirmatory execution is separated from protocol development. The canonical manifest binds task bundles, split hashes, configuration, runner code, model identity, and source result tables. Pilot namespaces use disjoint task seeds. Once a confirmatory protocol is frozen, a change requires an explicit amendment and replacement of the affected cells rather than silent in-place repair.

The canonical table incorporates four documented corrections. M04, M05, and M08 use strictly future definitions that satisfy their construction contracts. M10's strict mask retains its legitimate injected field. MI uses a fixed random state on training data for the affected diagnostic cells, and natural-case preprocessing is training-fitted. Superseded rows are retained for provenance but excluded from canonical claim generation. These amendments narrow the evidence to the final protocol and prevent earlier, semantically different outputs from being averaged into the paper results.

3.7 Statistical Analysis and Claim Scope

The controlled task is the independent unit. Core measurement intervals use a 20,000-draw hierarchical bootstrap that samples tasks and then injection seeds. Mechanism tests use task-level sign flips with Holm correction over the eleven mechanisms. The three category contrasts form a separate Holm family. The pre-specified simple-minus-structured contrast receives the core binary support decision; individual mechanism profiles remain descriptive.

For clarity, let $\mathcal{V}$ denote the learner set, and let $\mathcal{M}_{\text{si}}$ and $\mathcal{M}_{\text{st}}$ denote the simple and structured mechanism families. The category-level harm for task t and seed s is

$$\bar{X}_{t,s,c} = \frac{1}{|\mathcal{V}| \|\mathcal{M}_c\| |\mathcal{A}|} \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}_c} \sum_{a \in \mathcal{A}} X_{(t,m,a,s),t}, \quad (13)$$

so the prespecified contrast is

$$\hat{\Gamma}_{\text{si-st}} = \frac{1}{|\mathcal{T}| |\mathcal{S}|} \sum_{t \in \mathcal{T}} \sum_{s \in \mathcal{S}} (\bar{X}_{t,s,\text{si}} - \bar{X}_{t,s,\text{st}}). \quad (14)$$

Writing the seed-averaged paired contrast as e_t , its exact two-sided task-level sign-flip value is

$$p_{\text{flip}} = 2^{-|\mathcal{T}|} \sum_{\epsilon \in \{-1,1\}^{|\mathcal{T}|}} \mathbf{1} \left[\left| \frac{1}{|\mathcal{T}|} \sum_t \epsilon_t e_t \right| \geq \left| \frac{1}{|\mathcal{T}|} \sum_t e_t \right| \right]. \quad (15)$$

For core confidence intervals, each bootstrap draw samples 20 tasks with replacement and then samples five injection seeds

with replacement inside each selected task. All strengths, mechanisms, and learners associated with a drawn task-seed pair remain together. This preserves the pairing that defines X_{qt} and prevents the 27,500 model cells from acting as independent replicates. The simple-minus-structured interval is the 2.5th and 97.5th percentiles of 20,000 hierarchical draws. Its exact sign-flip test first averages seeds within task, enumerates all 2^{20} task-sign assignments, and uses the two-sided absolute mean. Holm adjustment is applied jointly to the three declared family contrasts. The eleven mechanism tests form a different Holm family; their profiles remain descriptive regardless of adjusted value.

Governance differences are first averaged within each of the 20 controlled tasks. With $\mathcal{M}$, $\mathcal{A}$, and $\mathcal{S}$ denoting the mechanism, strength, and injection-seed sets, the task-level paired effect is

$$d_{t,p} = \frac{1}{|\mathcal{M}||\mathcal{A}||\mathcal{S}|} \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}} \sum_{s \in \mathcal{S}} (\text{SDR}_{q,P3,k_q(p)} - \text{SDR}_{q,P2,k_q(p)}), \quad (16)$$

and the registry-level estimand is the equal-task average

$$\hat{\Delta}_p = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d_{t,p}. \quad (17)$$

For a mechanism family c with member set $\mathcal{M}_c$, replacing $\mathcal{M}$ by $\mathcal{M}_c$ in Eq. 16 gives

$$\hat{\Delta}_{p,c} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d_{t,p,c}. \quad (18)$$

We assess these estimands with a paired 10,000-draw cluster bootstrap over tasks. For draw b , tasks are sampled with replacement and the complete task effect is carried with each sampled index:

$$\hat{\Delta}_p^{*(b)} = \frac{1}{|\mathcal{T}|} \sum_{i=1}^{|\mathcal{T}|} d_{T_i^{*(b)},p}, \quad T_i^{*(b)} \stackrel{\text{iid}}{\sim} \text{Unif}(\mathcal{T}). \quad (19)$$

We report the observed mean task difference, percentile interval, and the fraction of bootstrap draws above zero:

$$\begin{aligned} \text{CI}_{0.95} &= [\mathcal{Q}_{0.025}(\hat{\Delta}_p^*), \mathcal{Q}_{0.975}(\hat{\Delta}_p^*)], \\ \hat{P}_{P3>P2} &= \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\hat{\Delta}_p^{*(b)} > 0]. \end{aligned} \quad (20)$$

The latter is a bootstrap superiority probability, not a p -value. Because tasks form a designed registry rather than a sample from a defined task population, all intervals quantify stability to registry reweighting. They do not support population inference to arbitrary tabular datasets.

4 Measurement Results

4.1 Simple Contamination Produces More Harm on Average

The prespecified simple-minus-structured contrast in paired AUROC harm was 0.164 (95% hierarchical bootstrap CI [0.150, 0.178]; Holm-adjusted exact task-level sign-flip $p = 5.72e - 06$). The direction was positive across all five core learner families. This category average does not imply that every structured mechanism is harmless: M09 is a structured counterexample with mean harm 0.235.

The learner-family consistency is descriptive rather than a second support decision. LR, bagged trees, boosted trees, and

TabM differ in their ability to represent interactions and partitions, but each yielded a positive simple-minus-structured average. This observation reduces the likelihood that the pooled direction is produced by one learner family alone. It does not show that learner capacity causes the category contrast, because hyperparameters, regularization, and representation are fixed components of the named model implementations rather than randomized treatments.

The mechanism rows further qualify the category summary. M01 and M02 show the expected vulnerability to explicit target proxies, while redundant M06 and mixed M10 yield still larger mean harm. M04 and M05 are construction-invalid despite estimates close to zero: their future-window values can be weakly useful in this particular generator and representation. M09 demonstrates the opposite possibility. Its source categories are structurally defined, yet the complete one-hot representation makes the source shift exploitable. Therefore “structured” describes how invalid information is organized, not a guarantee of low statistical association or low downstream harm.

4.2 Detectability and Exploitability Form Crossed Profiles

Mechanism profiles show why a single leakage severity score is insufficient. M03 combines low primary-MI detectability ($D=0.079$) with substantial harm ($X=0.217$). M08 has intermediate detectability ($D=0.356$) and an imprecise small mean harm ($X=0.003$, CI spanning zero). M09 is both more detectable and strongly exploitable in the fixed one-hot source representation. These rows are descriptive profiles, not post hoc thresholded claims.

The diagnostic sensitivity suite makes the conditioning on D concrete. Across all mechanisms, the aggregate ordering of the four diagnostics differs from their ordering on individual mechanisms. M03 is the clearest example: its normalized AP ranges from 0.035 under absolute correlation to 0.602 under RF permutation importance, whereas primary MI yields 0.079. M04 and M05 remain weakly localized under all four evaluated diagnostics. These comparisons show that “hard to detect” must name a score family and representation. They do not establish RF permutation as a universally superior detector, because the method comparison lacks a frozen simultaneous paired interval and the best-per-mechanism choice uses oracle labels retrospectively.

At the eleven-mechanism level, D and X have a positive descriptive rank association, but category membership explains much of the apparent relation and leave-one-mechanism-out checks do not support a portable prediction rule. This is why Fig. 2 is a profile map rather than a fitted calibration curve. The benchmark can identify mechanisms for which a diagnostic and a learner co-vary; it cannot infer exploitability for an unseen mechanism from D alone.

4.3 Natural Cases Broaden Context but Not the Sampling Frame

Five selected real-data case studies provide boundary-specific consistency checks (Table 2): UCI Bank Marketing [17], a local Lending Club mirror, U.S. flight delays [18], Chicago food inspections [2], and NYC 311 [3]. All have positive mean paired harm, but their size, feature representation, and contamination labels differ. The mean over these five selected tasks is 0.273. We do not use these cases for population inference.

The five case means span a wide range rather than forming a homogeneous replication set. Flight and inspection boundaries

Table 1: Measurement profiles for the frozen mechanism registry. Harm is permissive-minus-strict test AUROC. D is normalized average precision for the primary mutual-information ranker. Harm intervals are task-hierarchical 95% intervals. Mechanism rows are descriptive; the prespecified simple-minus-structured contrast is reported in the text.

ID	Mechanism	Family	Harm	95% CI	D
M01	Direct Target Copy	simple	+0.180	[+0.164, +0.196]	0.844
M02	Noisy Target Proxy	simple	+0.204	[+0.186, +0.222]	0.918
M03	Nonlinear Target Transform	boundary	+0.217	[+0.201, +0.234]	0.079
M04	Post-Outcome Aggregation	structured	-0.001	[-0.004, +0.001]	0.081
M05	Temporal Look-Ahead	structured	-0.001	[-0.003, +0.001]	0.084
M06	Redundant Leakage Cluster	simple	+0.284	[+0.261, +0.306]	0.998
M07	Sparse Subgroup Leakage	boundary	+0.124	[+0.114, +0.133]	0.813
M08	Entity Leakage	structured	+0.003	[-0.002, +0.009]	0.356
M09	Source Leakage	structured	+0.235	[+0.216, +0.254]	0.541
M10	Mixed Leakage	simple	+0.226	[+0.206, +0.245]	0.944
M11	Graph-Mediated Leakage	boundary	+0.286	[+0.263, +0.309]	0.829

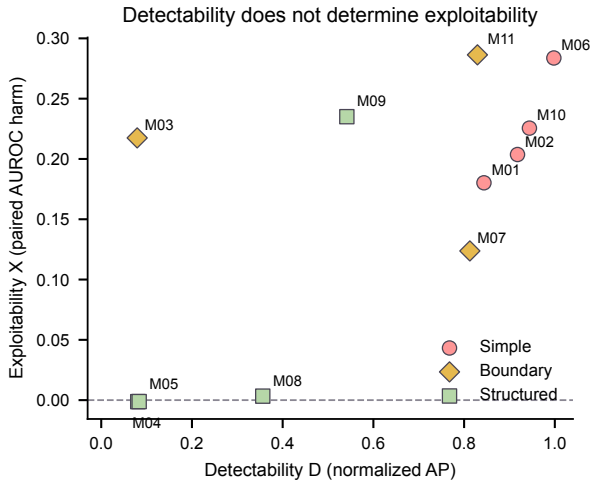


Figure 2: C/D/X measurement profiles. Each point is one mechanism mean over the frozen registry. D is normalized average precision under the primary training side mutual information diagnostic; X is paired AUROC harm. Shape and color encode construction family. The figure is descriptive and does not fit a deployment rule.

show large observed harm, whereas the NYC 311 boundary is close to the strict reference. This spread is substantively useful: it shows that declaring a field unavailable does not determine the magnitude of model exploitation. However, the tasks differ in sample construction, table width, encoded representation, unavailable-field count, and application semantics. Averaging them provides a compact audit summary, not a transport estimate. The fact that all five observed means are positive is reported as a fixed-case description and is not promoted to a population claim.

The matched-cost governance sensitivity is also heterogeneous. The descriptive mean P3-minus-mean-P2 SDR across the five cases is +0.187 (case-bootstrap interval [+0.025, +0.323]), but the five cases do not form a population sample. Bank Marketing (+0.102), Lending Club (+0.266), BTS Flights (+0.348), and Chicago Food (+0.326) favor P3, whereas NYC311 favors P2 (-0.108). The exact two-sided sign-flip value over the five fixed

case effects is 0.1875. We therefore classify natural governance as mixed descriptive evidence rather than external confirmation.

5 Governance Results

5.1 Blind MI Outperforms Random Removal Across Budgets

At a 20% encoded-column budget, P3 exceeded the within-key mean of 20 P2 realizations by +0.043 SDR for LR (95% paired dataset-cluster bootstrap CI [+0.004, +0.078]), +0.055 for RF ([+0.025, +0.082]), and +0.056 for LightGBM ([+0.028, +0.082]). Direct paired learner contrasts all cross zero, so the supported statement is that the direction and magnitude are similar in this three-learner panel, not that the effects are equivalent or learner-invariant. The encoded-column budget curve additionally shows the recall-retention trade-off across 1%, 5%, 10%, and 20% removal.

Random removal is a meaningful negative control in this setting because feature deletion can improve SDR without locating contamination. A permissive model may overfit weak legitimate fields, and deleting any such field can move its AUROC toward the strict score. P3 must therefore exceed the within-key mean P2 result, not merely obtain positive SDR. The observed advantage at all four nominal budgets indicates that MI ranking contributes information beyond deletion count in this LR panel.

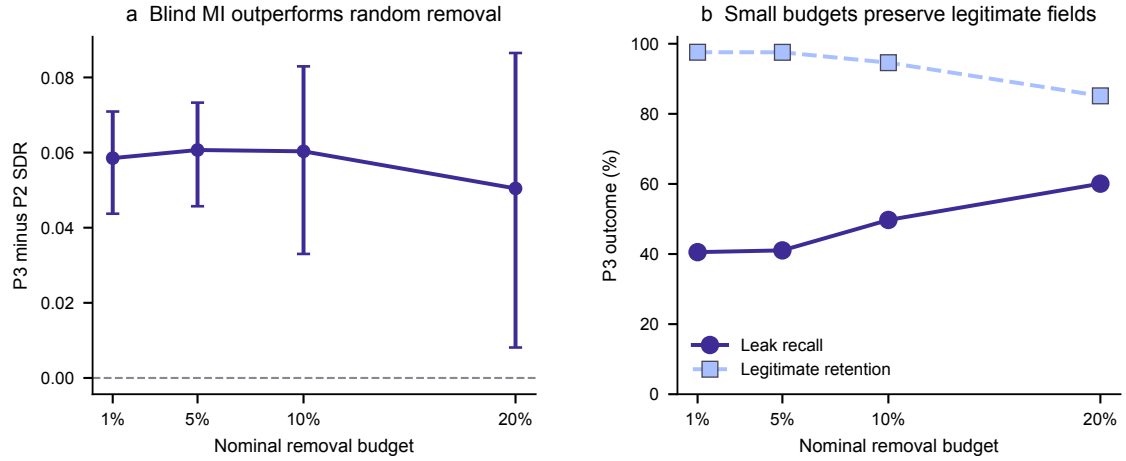
The budget curve should not be interpreted as four fully independent policy experiments. On narrow tasks, adjacent fractions can map to the same rounded integer $k_q(p)$ and therefore reuse one fitted result under multiple nominal labels. The 5,500 task keys are complete at each nominal fraction, but the number of distinct interventions varies by task width. Figure 3 accordingly summarizes operational budget labels; it does not fit a smooth dose-response model.

5.2 The Structured Average Hides Repair-Opportunity Heterogeneity

At a 20% budget, the P3-P2 difference was +0.109 for simple mechanisms (CI [+0.060, +0.153]) and +0.044 for boundary mechanisms (CI [+0.002, +0.079]). For structured mechanisms it was -0.003 (CI [-0.039, +0.027]). The structured interval crosses zero, so the supported statement is that this experiment finds no reliable P3 advantage for that family. It does not establish equivalence between P3 and P2 or a general failure of MI on structured leakage. M04, M05, and M08 have little initial strict-full

Table 2: Boundary-specific real-data case studies. These selected tasks provide descriptive consistency checks, not population inference.

Task	Declared prediction boundary	Features	Leak fields	Mean harm
BankMarketing	Before call completion	20	1	+0.131
LendingClub	At loan origination	140	29	+0.319
BTSFlights	Before scheduled departure	108	72	+0.470
ChicagoFood	Before inspection outcome	16	2	+0.430
NYC311	At complaint creation	40	2	+0.013

**Figure 3: Budget sensitivity of blind MI governance. a, observed P3 minus P2 SDR with paired dataset-cluster bootstrap 95% intervals over 20 controlled tasks. Positive values favor P3. b, contaminated field recall and legitimate-field retention for P3. P2 and P3 remove the same number of encoded fields at each task and nominal budget.**

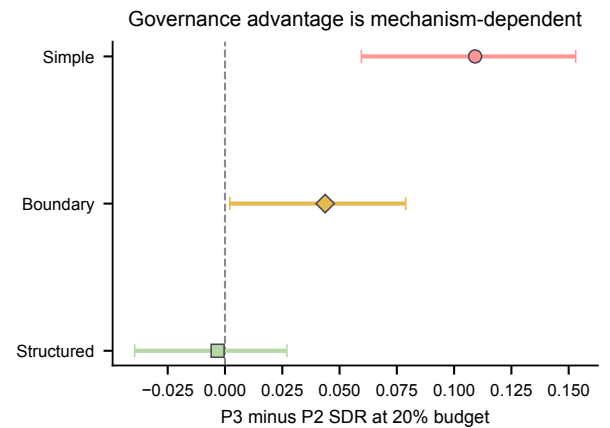
gap, whereas M09 combines substantial distortion with a positive P3–P2 contrast across all three evaluated learners.

The family decomposition is a direct effect comparison, not an inference from one interval being above zero and another crossing zero. Each family estimate recomputes the paired P3–P2 difference within its declared mechanism set and then clusters over tasks. The positive simple and boundary estimates therefore identify where the aggregate gain is concentrated. The structured estimate instead averages mechanisms with different repair opportunity and should not be treated as a category-level capability boundary.

5.3 Semantic-Group Cost Narrows the Overall Claim

The primary protocol charges encoded columns, so removing the complete M09 source field costs eight units. A frozen sensitivity instead treats its eight one-hot indicators as one semantic group, scores each group by the maximum training-side MI of its members, and samples P2 uniformly over groups. M09 is the only mechanism whose frozen representation expands one semantic source into several encoded columns; the remaining 5,000 keys therefore retain their encoded-cost results in the 5,500-key re-composition.

M09 remains positive under group cost: its P3–P2 contrast changes from +0.149 (95% CI [+0.125, +0.171]) to +0.109 ([+0.054, +0.158]). The paired semantic-minus-encoded change is -0.040 ([−0.076, −0.008]). After recomposition, the overall estimate is +0.040 with an interval that crosses zero ([−0.003, +0.077]), compared with

**Figure 4: Mechanism-family heterogeneity at a 20% budget. Points show observed P3 minus P2 SDR and bars show paired dataset-cluster bootstrap 95% intervals. Marker shape and color redundantly encode family. The structured estimate is a negative result, not an equivalence claim.**

+0.043 ([+0.004, +0.078]) under encoded cost. Thus M09 is not an artifact of charging eight units, but the aggregate governance claim is sensitive to the representation-cost definition.

Table 3: Matched-cost governance results. Differences are P3 minus P2 strict-distance reduction (SDR); positive values favor blind MI. Probability is the fraction of paired dataset-cluster bootstrap draws above zero, not a p -value. Recall and retention describe P3.

Scope	Budget	P3–P2 SDR	95% CI	$\Pr_{boot}(\Delta > 0)$	Leak recall	Legit. retention
All Mechanisms	1%	+0.059	[+0.044, +0.071]	1.000	40.5%	97.6%
All Mechanisms	5%	+0.061	[+0.046, +0.073]	1.000	41.0%	97.6%
All Mechanisms	10%	+0.060	[+0.033, +0.083]	1.000	49.7%	94.6%
All Mechanisms	20%	+0.050	[+0.008, +0.087]	0.989	60.1%	85.1%
Simple	20%	+0.109	[+0.060, +0.153]	1.000	92.9%	86.7%
Boundary	20%	+0.044	[+0.002, +0.079]	0.980	56.2%	85.7%
Structured	20%	-0.003	[-0.039, +0.027]	0.449	30.2%	83.1%

5.4 From Measurement to Action

The measurement and governance results align at the family level without establishing a universal transfer rule. Simple contamination is usually highly detectable, strongly exploited, and amenable to top-ranked field removal. Boundary mechanisms are heterogeneous but retain a positive aggregate P3 advantage. Structured mechanisms encode invalidity through time, entity, and source relationships that a univariate training side score does not represent uniformly. This interpretation is consistent with the observed profiles and policy outcomes, but it is not a causal decomposition of policy performance.

The crossed profiles also warn against treating high D as sufficient for governance. M09 is detectable and harmful under its fixed representation, yet structured governance as a family does not show a reliable advantage. A field ranking can identify association while still removing legitimate correlates or failing to represent the semantic group that should be governed. The benchmark therefore evaluates localization, downstream distance, recall, and retention as separate outcomes.

The policy results also separate identification from repair. A contaminated column counted in recall may be statistically redundant with another retained column, so removing it need not change AUROC. Conversely, removing a legitimate correlate can accidentally offset permissive inflation and improve SDR while reducing retention. The joint outcome vector (SDR, Rec, Ret) prevents either case from being reported as unqualified success. In the present results, low-budget P3 combines a positive paired advantage with high legitimate retention, whereas larger budgets buy additional recall at a visible retention cost.

6 Related Work

Leakage semantics. Classical leakage formulations distinguish information that is available when a model is developed from information that will be available when its prediction is consumed [9]. This semantic perspective is stronger than a correlation rule: an aggressively predictive field may be legitimate, while a weak future-derived field remains invalid. Recent work links leakage to reproducibility in ML-based science [8] and formalizes domain-specific label leakage in health care [4]. Empirical studies in connectome prediction further show that leakage can inflate reported performance under otherwise familiar modeling pipelines [14]. LeakBench-Tab adopts the same boundary-first principle, but turns it into a factorial measurement contract in which validity, localization, and exploitation are separate variables.

Detection at code, data, and representation levels. Static program analysis targets leakage pathways visible in data-science

code, such as fitting preprocessing outside the training split [16]. Contamination audits in language benchmarks instead often ask whether evaluation content was memorized and subsequently exploited [11]. Feature-level screening occupies another level: it ranks columns from observed training data but typically cannot determine whether their information is temporally or operationally available. These approaches are complementary because they observe different evidence. Code analysis can reveal an invalid fit scope without ranking columns; a statistical ranker can reveal a target proxy without knowing its lineage; and a semantic contract can identify a future field even when its sample association is weak. Our C/D/X decomposition makes these non-equivalences explicit rather than combining them into a single detector score.

Tabular evaluation under controlled shifts. TableShift studies tabular prediction under distribution shifts, while broad comparative benchmarks examine when boosted trees, bagged trees, linear models, and neural architectures succeed [5, 7, 12]. Those resources vary datasets, domains, or learner families to characterize predictive performance. LeakBench-Tab holds a task instance and split fixed while perturbing the information boundary through auditable mechanisms. Learner diversity is used to measure exploitability conditional on representation, not to declare a new winner among logistic regression, random forests, boosting, and TabM [1, 6, 10, 13]. This focus also explains why the natural cases are treated as boundary audits rather than merged with the controlled panel.

Feature selection versus leakage governance. Mutual information, linear coefficients, correlation, and permutation importance measure predictive relevance under different function classes. None is intrinsically a semantic validity test. Conventional feature selection usually optimizes predictive performance or sparsity; leakage governance asks whether a constrained intervention moves an evaluation toward a boundary-valid reference while preserving legitimate information. Our P2/P3 comparison therefore matches removal cost, records contaminated-field recall and legitimate-field retention, and evaluates downstream strict distance. The scientific contribution lies in the observed regime map and negative structured result, not in presenting MI ranking as a new selection algorithm.

7 Discussion

7.1 What the Evaluation Changes

Leakage audits often end after declaring suspicious fields or measuring an inflated score. Our results show that an actionable evaluation needs at least four linked but distinct objects: the

semantic contamination label, a blind localization score, downstream exploitability, and the effect of a removal policy under explicit cost. Reporting only one object hides important cases. Low D does not validate a field, high X does not reveal which field caused the inflation, and high D does not guarantee that removal will preserve legitimate utility.

Matched-cost evaluation is central to this conclusion. P3 is not compared with a baseline that removes fewer fields, and the evaluation records retention as well as recall. The 1% result is practically informative because it combines a positive distance reduction with high legitimate retention. The 20% result shows the opposite side of the trade-off: recall rises, but retention falls.

7.2 Implications for Leakage Audits

The findings suggest an audit sequence rather than an automatic deletion rule. An analyst should first declare the prediction event and label each field by availability or lineage. Blind scores can then prioritize manual review, but their rankings should be evaluated against a held-back semantic mask when a benchmark is available. Finally, any removal policy should be compared with an equal-cost negative control and assessed on downstream behavior. This sequence prevents a high association score from silently becoming a validity judgment.

Budget choice is part of the governance decision. At small budgets, MI removal preserves most legitimate fields and can cheaply eliminate direct proxies. At larger budgets, the marginal recall gain is purchased with additional legitimate-field deletion. A deployment process would therefore need a domain-specific cost function over missed contamination, lost legitimate signal, calibration, and decision utility. The benchmark exposes ingredients for that choice but deliberately does not collapse them into a universal scalar utility.

The semantic-group sensitivity shows why cost definition must be part of the audit contract. Grouping M09's indicators preserves its positive effect but widens the full-panel uncertainty enough to cross zero. Richer policies could add feature-lineage constraints or treat temporal availability and entity aggregation as first-class metadata. Their evaluation should preserve the same oracle-isolation rule: metadata available to the policy may be used, but the hidden contamination label should remain an evaluation object.

7.3 Alternative Explanations and Required Follow-Ups

The observed family pattern can have several compatible explanations. Simple mechanisms may be easier for MI to rank because their invalid signal is carried by individual encoded columns. Structured mechanisms may require group or conditional statistics, but they may also exhibit weaker exploitability in the particular generators, leaving less inflation for any policy to repair. Learner representation adds a third possibility, although the similar LR, RF, and LightGBM contrasts do not reveal a detectable learner interaction in the present panel. The current factorial design describes the joint outcome of these factors; it does not identify their separate causal contributions.

The largest remaining ambiguity concerns policies that use lineage, temporal, or entity metadata rather than blind univariate association. These policies would test whether low-gap structured mechanisms become governable when the intervention represents their construction semantics. Natural evaluation also

remains narrow: five fixed cases cannot establish a population-level transport claim, and the negative NYC311 effect requires application-specific explanation rather than averaging away.

7.4 Limitations

The controlled tasks are synthetic panel generators, not twenty independent real-world datasets. Their bootstrap intervals describe registry stability, not a task population. The governance experiment covers LR, RF, and LightGBM, but these named implementations do not establish invariance across model families or tuning choices. The direct learner-interaction intervals cross zero; they are tests of detected difference, not equivalence tests.

P2 uses 20 frozen random field selections for each task key and budget, and the primary estimand averages P2 within key before task-cluster resampling. This integrates the declared random-policy variance but does not cover every possible randomization scheme. Nominal budget labels can also map to the same integer removal count on tasks with few fields, so adjacent budget points are not fully distinct interventions for every task.

The primary blind policy is univariate mutual information. It does not encode lineage, temporal availability, entity grouping, or source semantics. The structured-family average should not be generalized to all metadata-aware or group-aware policies. Conversely, P3's aggregate advantage should not be read as evidence that association determines validity.

The five real-data audits are selected, boundary-specific case studies. One local Lending Club mirror cannot be redistributed until its source and license are verified. AUROC distance does not replace calibration, decision utility, or operational monitoring. Runtime and resource overhead are also outside the current frozen evidence and should be added before making systems-efficiency claims.

7.5 Threats to Validity

Construct validity. SDR measures movement toward a frozen strict reference, not the business value of a deployed decision. Its interpretation depends on the correctness of each declared prediction boundary and mechanism contract. Recall and retention use the contamination mask only after policy selection and are therefore evaluation measures rather than deployable observables. Encoded-column recall also differs from semantic-field recall when categorical fields expand into multiple indicators. The benchmark states the encoded unit explicitly, but a production policy may require group-level metrics.

Internal validity. The paired protocol holds splits, task content, model seeds, and removal cost fixed within each comparison. Bundle, split, and runner hashes reduce implementation drift, but they cannot rule out all library-specific behavior. Hyperparameters are fixed named implementations rather than optimized independently for every mechanism; this improves comparability but may understate the response of a specially tuned learner. Protocol amendments are provenance-bound and replace affected cells, yet their necessity also illustrates how semantic bugs can survive ordinary completion checks.

External and statistical validity. The designed registry deliberately varies mechanism structure, but it does not define a sampling frame of tabular applications. Mechanism-family intervals and bootstrap probabilities should be read as stability summaries for this registry. The natural cases broaden the contexts inspected but remain selected case studies; their governance result is mixed and descriptive. Multiplicity correction protects the

declared contrast families, but descriptive mechanism profiles and diagnostic sensitivity rows should not be mined as independent confirmatory discoveries.

8 Reproducibility and Artifact Boundary

The release binds the canonical 27,500-cell measurement table, paper claim state, governance runner, 55,000-cell policy ledger, bootstrap analysis, and paper-facing CSV tables by SHA-256. The release validator deep-runs statistical tables and reports 233 passing tests with no failures. Superseded runners and intermediate cluster summaries are retained for provenance but excluded from claim generation. Paper figures and tables are regenerated from three compact CSV assets: the mechanism results, governance results, and natural cases.

Reproduction is organized in three layers. The *paper layer* regenerates all numeric macros, tables, and figures from the three compact CSV assets and records their hashes in a paper-artifact manifest. The *analysis layer* rebuilds measurement and governance summaries from canonical cell ledgers, including the task-level bootstrap objects. The *execution layer* reruns frozen task bundles through named model and policy runners after verifying bundle, split, configuration, and code identities. A reader can therefore inspect the paper without rerunning models, recompute claims without rebuilding tasks, or execute the complete experiment when the required packages and hardware are available.

Claim state is machine-readable and distinct from prose. Each central claim records an estimand, evidence tier, allowed wording, prohibited wording, and scope limitation. The paper generator does not infer support from arbitrary tables; it consumes the frozen paper-facing assets. This separation is intended to prevent a later figure edit, exploratory table, or superseded run from silently changing a conclusion.

The public artifact and reproduction instructions are maintained at <https://github.com/lzzzhh/LeakBench-Tab>. The paper-facing artifact does not redistribute the local-only Lending Club mirror. All non-oracle policy selectors can be audited without access to the contamination mask; the mask is used only for evaluation after selection.

9 Conclusion

LeakBench-Tab connects leakage measurement to matched budget action. Across a frozen controlled registry, blind training-side mutual information removal has a positive 20% encoded-cost P3–P2 interval for LR, RF, and LightGBM. The structured family average is not a failure boundary: it mixes low-gap mechanisms with the positive M09 counterexample. Semantic grouping preserves the M09 advantage but causes the recomposed overall interval to cross zero, and natural governance is positive in four of five fixed cases. The central lesson is therefore conditional: detectability can guide repair when distortion is both present and localizable, but the conclusion depends on cost representation and application context.

Leakage benchmarks should separate invalidity, localization, exploitation, and repair. These quantities coincide for direct

proxies but diverge for interactions, future aggregates, entities, sources, and redundant groups. A defensible result therefore requires a semantic reference, oracle isolation, matched cost, paired downstream evaluation, and task-level uncertainty. The release provides these components for the present scope. Next experiments should test lineage-, time-, and entity-aware policies and explain the negative NYC311 governance result under its application boundary.

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